higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE

## POWER MACHINES N5

## 29 NOVEMBER 2019

**This marking guideline consists of 9 pages.**

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## QUESTION 1

1.1
- Must be designed and manufactured according to the code of practice
- Must be approved by the inspector if no code exists for such a boiler
- Must be manufactured under supervision of an inspection authority (3 × 2) (6)

1.2
- More work done is achieved due to greater expansion of steam.
- Water in the circuit can be treated to reduce scale formation in the boiler.
- The pressure can be operated below atmospheric pressure. (Any 2 × 2) (4)

**[10]**

## QUESTION 2

$V_1 = 0{,}15\ m^3$ $T_1 = 40\ °C$
$P_1 = 310\ kPa$ $m = 0{,}5\ kg$
$P_2 = 680\ kPa$ $C_v = 0{,}718\ kJ/kg.K$

2.1
$$P_1V_1 = mRT_1$$
$$R = \frac{P_1V_1}{mT_1}$$
$$= \frac{310 \times 0.15}{0.5 \times 313}$$
$$= 0,297\ kJ/kg.K$$
(3)

2.2
$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}}$$
$$T_2 = T_1\left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}}$$
$$= 313\ \left(\frac{680}{310}\right)^{\frac{1.35-1}{1.35}}$$
$$= 383,698\quad K$$
(3)

2.3 $P_2V_2 = mRT_2$

$$V_2 = \frac{mRT_2}{P_2}$$

$$= \frac{0{,}5 \times 0.297 \times 383{,}698}{680}$$

$$= 0{,}084\, m^3$$

*OR*

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{n-1}$$

$$V_2 = \frac{V_1}{\left(\frac{T_2}{T_1}\right)^{\frac{1}{n-1}}}$$

$$= \frac{0{,}15}{\left(\frac{383{,}698}{313}\right)^{\frac{1}{1.35-1}}}$$

$$= 0{,}084\ m^3$$ (3)

2.4 $$WD = \frac{P_1V_1 - P_2V_2}{n-1}$$

$$= \frac{(310 \times 0.15) - (680 \times 0{,}084\ )}{1{,}35 - 1}$$

$$= -30{,}343\ kJ$$

*OR*

$$WD = \frac{mR\,(T_1 - T_2)}{n-1}$$

$$= \frac{0.5 \times 0{,}297\,(313 - 383\,,698\ )}{1{,}35 - 1}$$

$$= -29\,,996\ kJ$$ (4)

2.5 $$\Delta U = mC_v(T_2 - T_1)$$

$$= (0{,}5)(0\,,718)(313 - 383\,,698)$$

$$= -25\,,381\ \text{kJ}$$ (3)

2.6 $$Q = \Delta U + WD$$ ✓

$$= -25\,,381 + (-29\,,996)$$ ✓

$$= -55\,,377\ \text{kJ}$$ ✓✓ (4)

**[20]**

## QUESTION 3

3.1
$$x = \frac{M}{M+m}$$
$$= \frac{1,5}{1,5+0,15}\checkmark$$
$$= 0,909\checkmark$$
(3)

3.2
$$P = \left(B_m + M_m\right)\frac{101,325}{760}$$
$$= \left(775+652\right)\frac{101,325}{760}$$
$$= 190,25 \text{ kPa}\checkmark$$
$$\approx 190 \text{ kPa}$$

At $P = 190$ kPa
$h_f = 498$ kJ/kg ✓
$h_{fg} = 2\ 206$ kJ/kg ✓

At 120 kPa, $t_{ss} = 110\ ^\circ C$ ✓
$h_g = 2\ 683$ kJ/kg ✓
$t_s = 104,8\ ^o C$ √ 3

+

*Enthalpy before throttling = Enthalpy after throttling*
$$h_1 \text{ at } 190\ kPa = h_2 \text{ at } 120\ kPa$$
$$h_f + xh_{fg} = h_g + c_p(t_{ss} - t_s)$$
$$498 + x(2206) = 2683 + 2,1(110 - 104,8)$$
$$x = 0,995\checkmark$$
(9)

3.3
$$x = x_1 \times x_2\checkmark$$
$$x = 0,909 \times 0,995$$
$$x = 0,904$$
(4)

3.4 $V = xV_g$

$V = 0,904 \times 0,929$

$V = 0,8398 \frac{m^3}{\text{kg}} \times 3 \text{ kg}$

$V = 2,519 \ m^3$ (4)

**[20]**

## QUESTION 4

4.1 4.1.1

$$p = (B_m - M_m)\frac{101{,}325}{760}$$
$$= (760-660)\frac{101{,}325}{760} \checkmark$$
$$= 13{,}332\ kPa \checkmark$$
$$\approx 13\ kPa$$

(2)

4.1.2 At $P$ = 13 kPa $h_f = 214$ kJ/kg $h_{fg} = 2\,380$ kJ/kg
$h_c = 192$ kJ/kg at $t_c = 45{,}8\ ^\circ C$

heat emitted by steam = heat absorbed by water

$h_1$ At 13 kPa = $h_2$

$$m_s\left[h_f + xh_{fg} - h_c\right] = mc_p\Delta T$$
$$8\,500\left[214 + (0{,}85 \times 2\,380) - 192\right] = m \times 4{,}2 \times (38-20)$$
$$m = 229\,927{,}249\ \frac{kg}{h} \checkmark$$

**ALTERNATIVE METHOD:**

At $P$ = 13 kPa $h_{fg} = 2\,380$ kJ/kg

$$h_c = c_p(t_{ss} - t_c)$$
$$h_c = 4{,}2(51{,}1 - 45{,}8)$$
$$h_c = 22{,}26\ kJ/kg$$

heat emitted by steam = heat absorbed by water

$$h_1 = h_2$$
$$m[xh_{fg} + h_c] = m \times c_p \times (t_2 - t_1)$$
$$8\,500[(0{,}85 \times 2380) + 22{,}26] = m \times 4{,}2 \times (38-20)$$
$$m = 229\,956{,}482\ \frac{kg}{h} \checkmark$$

(6)

4.1.3 At $t_c = 45,8\,^oC$, $P_s = 10$ kPa √

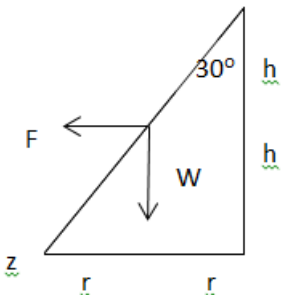
$$P_T = P_a + P_s$$
$$P_a = 13\,,\,332 - 10$$
$$= 3\,,\,332 \text{ kPa} \quad ✓$$

$$T_a = 273 + 45,8$$
$$= 318,8\,^oC \quad √$$

$$P_a V_a = mRT_a$$
$$V_a = \frac{mRT_a}{P_a}$$
$$= \frac{0,\,25 \times 0,\,3 \times 318,8}{3,332} \quad √ \quad √$$
$$= 7,176 \frac{m^3}{s} \times \frac{60\ s}{1\ \text{min}} \quad √$$
$$= 430,56\ m^3/\text{s} \quad ✓$$

(5)

4.2 $h = 0{,}28\cos 30^o$
$h = 0{,}242m$
$r = 0{,}28\sin 30^o$
$r = 0{,}14m$



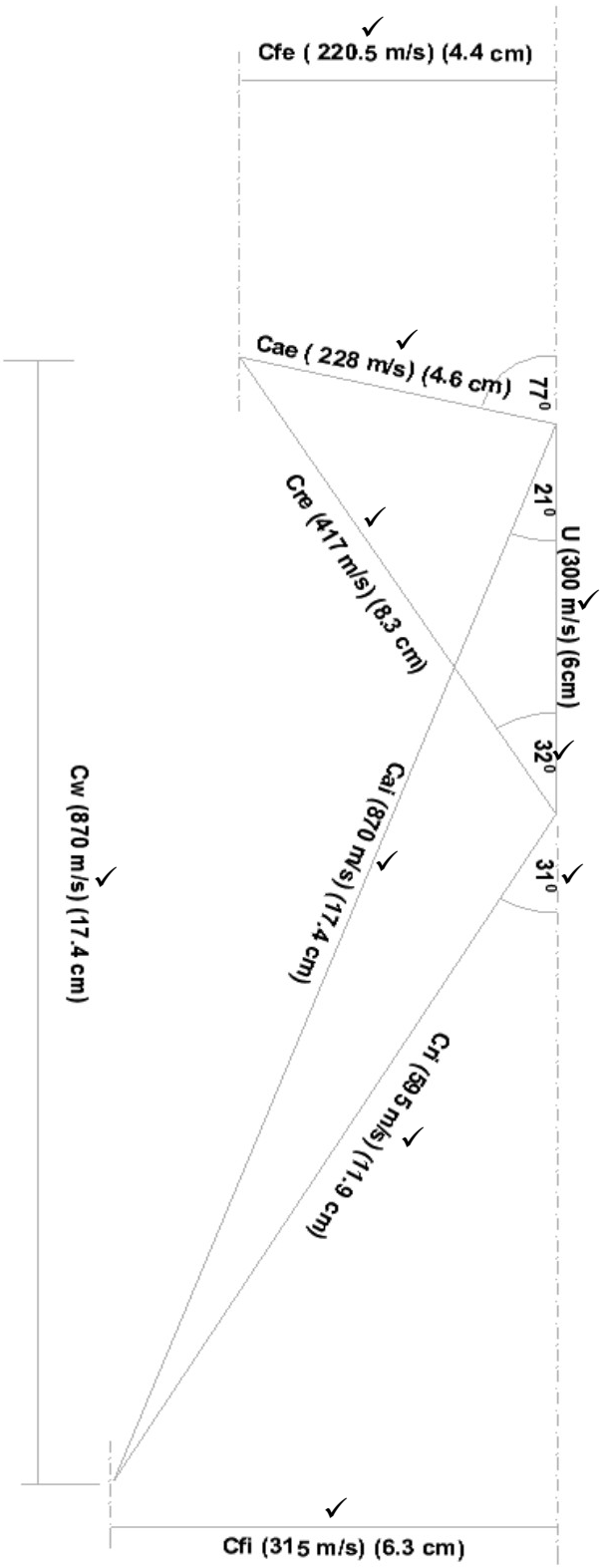

Take moments about z:

$$F \times h = W \times r$$
$$F \times 0,242 = (1,2 \times 9,81) \times 0,14$$
$$F = \frac{1,2 \times 9,81 \times 0,14}{0,242}$$
$$F = 6,81\ N$$

(7)
**[20]**

## QUESTION 5

5.1




(10)

5.2 5.2.1 $\alpha = 25^o$

5.2.2 $Cfe = 220,5$ m/s

(2 × 1) (2)

5.3
$$\eta = \frac{mU\left[(Cwi)-(Cwe)\right]}{Cai^2}$$
$$=\frac{2\times 300\times 870}{870^2}\times 100$$
$$= 68,965\%$$

(3)
**[15]**

## QUESTION 6

6.1
$$HCV=(CV_C.C)+CV_{H_2}.\left(H_2-\frac{O_2}{8}\right)+(CV_s.S)$$
$$HCV=(34\times 0,85)+(145)(0,07-\frac{0,02}{8})+(9\times 0,06)$$
$$HCV=39,228 \text{ MJ/kg}$$ ✓

(3)

6.2
$$LCV=HCV-h_{fg}(9H_2)$$
$$LCV=39,228-(2,424)(9\times 0,07)$$
$$LCV=37,701 \text{ MJ/kg}$$

(3)

6.3 $$M_{air} = \frac{100}{23}\left[\left(\frac{8}{3}\times 0{,}85\right)+\left(8\times 0{,}07\right)+0{,}06-0{,}02\right]$$

$$M_{air} = 12{,}463\ kg\ air/kg\ fuel$$

Mass of dry products:

$$CO_2 = 3{,}667\times 0{,}85$$
$$= 3{,}117 \text{ kg } CO_2/\text{kg fuel}$$
$$SO_2 = 2\times 0{,}06$$
$$= 0{,}12 \text{ kg } SO_2/\text{kg fuel}$$
$$N_2 = \frac{77}{100}\times 12{,}463$$
$$= 9{,}597 \text{ kg } N_2/\text{kg fuel}$$
$$\text{Total mass} = 3{,}117+0{,}12+9{,}597$$
$$= 12{,}894 \text{ kg products/kg fuel}$$

$$AHV = LCV - mC_p\left(t_2-t_1\right)$$
$$AHV = 37{,}701-\frac{12{,}894\times 1{,}045\times\left(230-36\right)}{1\ 000}$$
$$AHV = 37{,}701-2{,}614$$
$$AHV = 35{,}087 \text{ MJ/kg}$$

(9)
**[15]**

**TOTAL: 100**

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